



Review

Rejection sensitivity and romantic relationships: A systematic review and meta-analysis

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ABSTRACT

This meta-analysis explored whether rejection sensitivity relates to facets of romantic relationships. A comprehensive literature search identified 60 studies (147 effect sizes; 16,955 participants) that met inclusion criteria. Data were analysed using inverse-variance weighted random effects meta-analysis. Mean effect sizes from 21 meta-analyses provided evidence that more rejection sensitive individuals report lower levels of relationship satisfaction and relationship closeness, lower levels of perceived partner satisfaction, a greater likelihood of intimate partner violence (perpetration and victimisation), higher levels of relationship concerns and relationship conflict, and higher levels of jealousy and self-silencing behaviours. There was also some evidence that rejection sensitive individuals are more likely to engage in risky sexual behaviour and are more prone to sexual compulsivity. There was no evidence of publication bias and various levels of heterogeneity in computed averages. Random effects meta-regression identified participant age and sex as important moderators of pooled mean effects. These findings provide a foundation for theoretical development of rejection sensitivity in romantic relationships and should be of interest to relationship and marriage counsellors and other relationship professionals.

1. Introduction

The fundamental need to belong can often manifest as heightened sensitivity to cues of rejection and monitoring behaviour in response to external feedback (Baumeister & Leary, 1995). However, oversensitivity to rejection in benign social situations might adversely affect interpersonal relationships. Rejection sensitivity is the dispositional tendency to expect, easily perceive, and react intensely to rejection (Downey & Feldman, 1996). Rejection sensitivity falls broadly within the personality dimension of neuroticism (see Tackett & Lahey, 2017) and was first detailed and studied in the context of interpersonal and romantic relationships (Erikson, 1950). Since that time, much research has explored the role of rejection sensitivity on romantic relationship parameters. However, as far as we are aware, no previous attempt has been made to synthesise the full body of research. The current study sought to conduct a systematic review and meta-analysis of all research on rejection sensitivity and romantic relationship outcomes. We also sought to explore whether associations between rejection sensitivity and romantic relationship parameters are moderated by sociodemographic factors including age, sex, and world-region.

1.1. The rejection sensitivity model

The rejection sensitivity model (Downey & Feldman, 1996; Pietrzak et al., 2005; Romero-Canyas et al., 2010) describes rejection sensitivity as a cognitive-affective processing disposition to anxiously expect, readily perceive, and intensely react to rejection. The model incorporates three main components: anxious expectations, perceptions of rejection, and cognitive-affective reactions. Anxious expectations develop early in life, and often in response to unpleasant interactions in close relationships (e.g., conditional love by parents, family violence). The model also posits that people with high anxious expectations tend to perceive ambiguous behavioural cues as indicative of rejection. This is because such individuals are more attentive to rejection cues and tend to perceive such rejection cues as more threatening (Romero-Canyas et al., 2010). Reactions to perceived rejection often take the form of anger and hostility that can often elicit actual rejection and reinforce those initial anxious expectations (Pietrzak et al., 2005). Thus, rejection sensitivity has three main components: hypervigilance to rejection cues, an inability to differentiate between rejection cues and social cues, and the implementation of defensive mechanisms to tackle rejection cues

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(Pietrzak et al., 2005).

The rejection sensitivity model can also be considered through the lens of defence motivation (Pietrzak et al., 2005). The purpose of the defence motivation system is to respond to threats (in this case rejection cues) that might sever an individual's social ties (Romero-Canyas et al., 2010). However, anxious expectations detect misinterpreted rejection cues and provide false positive feedback to the defence motivation system. This triggers an overreaction of the defence motivation system inducing emotions and behaviours for self-protection. The overreaction increases the likelihood of actual rejection that, in turn, reinforces the initial anxious expectations (Romero-Canyas et al., 2010). In other words, this process reflects a self-fulfilling prophecy in which those individuals more fearful of rejection behave in ways that elicit rejection from others, thereby validating their initial concerns (Downey et al., 1999). The emotions and behaviours for self-protection that emerge in high rejection sensitive individuals is predicted to increase risk of maladaptive outcomes including depression, anxiety, personality disorders, body image disorders, loneliness, and self-harm (Pietrzak et al., 2005; Romero-Canyas et al., 2010).

1.2. Previous meta-analyses

Several meta-analyses have been conducted that test associations between rejection sensitivity and theoretical outcomes. A meta-analysis of 75 studies (Gao et al., 2017) found that rejection sensitivity was positively associated with mental health outcomes including depression, $k = 51$, $r = 0.32$ (95 % CI: 0.28, 0.35), anxiety, $k = 37$, $r = 0.49$ (95 % CI: 0.43, 0.55), borderline personality disorder, $k = 21$, $r = 0.47$ (95 % CI: 0.37, 0.56), body dysmorphic disorder, $k = 9$, $r = 0.46$ (95 % CI: 0.26, 0.66), and loneliness, $k = 11$, $r = 0.42$ (95 % CI: 0.38, 0.45). These associations corrected for publication bias and an important moderation effect was observed such that the association between rejection sensitivity and anxiety was weaker in samples with a greater percentage of women. Age, sex, and world-region did not moderate associations for other mental health outcomes (Gao et al., 2017). A subsequent meta-analysis of 43 studies focused exclusively on borderline personality disorder (Foxhall et al., 2019). This meta-analysis also found a positive association between rejection sensitivity and borderline personality disorder, $k = 31$, $r = 0.35$ (95 % CI: 0.31, 0.42), which remained significant after correcting for publication bias, $r_{\text{pseudo}} = 0.28$ (95 % CI: 0.23, 0.35). The association was not moderated by sample age, study design, or percentage of women in the sample (Foxhall et al., 2019).

A further meta-analysis of 39 studies also explored the association between rejection sensitivity and borderline personality disorder (Cavicchioli & Maffei, 2020). The meta-analysis found that people diagnosed with borderline personality disorder had higher levels of rejection sensitivity than normative samples, $k = 28$, $d = 0.85$ (95 % CI: 0.80, 0.91). There was some evidence of publication bias and substantial heterogeneity, that could not be explained by sample sex, sample size, or publication date (Cavicchioli & Maffei, 2020). A subsequent meta-analysis of 52 studies explored the contribution of rejection sensitivity to aggression and victimisation (Gao et al., 2021). Results showed that rejection sensitivity was positively associated with both aggression, $k = 23$, $r = 0.19$ (95 % CI: 0.13, 0.25), and victimisation, $k = 51$, $r = 0.38$ (95 % CI: 0.02, 0.33) (corrected for publication bias). The study also bifurcated aggression into reactive and proactive aggression. The relationship between aggression and rejection sensitivity was stronger for reactive, $k = 6$, $r = 0.23$ (95 % CI: 0.15, 0.32), compared to proactive, $k = 3$, $r = 0.09$ (95 % CI: -0.01 , 0.20) aggression.

1.3. The current study

The meta-analyses conducted to date support general predictions of the rejection sensitivity model (Downey & Feldman, 1996; Romero-Canyas et al., 2010), with rejection sensitive individuals reporting a greater prevalence of depression, anxiety, borderline personality

disorder, body dysmorphic disorder, loneliness, aggression, and victimisation. However, the original model prediction that rejection sensitive individuals are prone to negative relationship outcomes has not been synthesised using meta-analysis. A romantic relationship involves complex feelings in passionate and companionate forms, often involving a combination of emotional and sexual experiences (Diamond, 2004). Passionate forms are primarily driven by arousal that cause people to experience feelings of happiness, whereas companionate forms involve peacefully sustaining and committed bonds between two people (Diamond, 2004). In other words, there are many variables that fall under the umbrella of romantic relationships including relationship demographics (e.g., average relationship length), affective experiences (e.g., jealousy, relationship satisfaction), hostile behaviour (e.g., relationship conflict, intimate partner violence), and sexuality and sexual health (e.g., frequency of sexual activity, risky sexual behaviour).

The rejection sensitivity model predicts that rejection sensitivity will be important for affective, motivational, and behavioural relationship outcomes. Moreover, the model considers three facets of rejection sensitivity: anxious expectations, ready perceptions, and intense reactions to rejection. Anxious expectations cause individuals to avoid putting themselves in situations that involve potential rejection, and overestimate the possibility of rejection if they enter a romantic relationship (Pietrzak et al., 2005). Thus, the model predicts that rejection sensitive individuals will differ on relationship demographics, such as being less likely to enter a romantic relationship. Rejection sensitivity is also underpins an anxious or ambivalent attachment style (Romero-Canyas et al., 2010). Individuals with anxious or ambivalent attachment styles exhibit less closeness and trust in relationships, and therefore the model further predicts that rejection sensitivity should be negatively associated with affective relationship outcomes such as satisfaction, commitment, closeness, expression, and support (Downey & Feldman, 1996; Romero-Canyas et al., 2010).

Rejection-sensitive individuals with anxious or avoidant attachment styles also tend to have a negatively distorted perception of their relationship, do not form secure attachments with their partners, do not invest as much in their relationships, and negatively interpret actions of those around them (Normansell & Wisco, 2017). As a result, rejection sensitive individuals are predicted to display more hostile behaviours such as conflict, withdrawal, self-silencing, jealousy, and violence (Romero-Canyas et al., 2010). Moreover, the defence motivation system (whose real function is to respond to threats) overreacts to rejection cues and in turn induces emotions for self-protection. Thus, the model predicts that perception of rejection can trigger intense reactions in the form of jealousy, aggression and hostility among high rejection-sensitive individuals (Romero-Canyas et al., 2010). This hostility and aggression can elicit actual (and not just perceived) reactive rejection from their partners, presenting itself as a self-fulfilling prophecy (Romero-Canyas et al., 2010). In short, the model predicts that high rejection sensitivity will be associated with a greater likelihood of hostile behaviours including intimate partner violence.

Understanding the role of rejection sensitivity in romantic relationships is important given that romantic relationships are crucial in individual functioning and well-being (Mirsu-Paun & Oliver, 2017). The current study sought to synthesise the full body of research on rejection sensitivity and romantic relationships and test whether associations are moderated by sociodemographic factors. For instance, rejection sensitivity might be more or less important for relationship outcomes in particular world regions. Research has found that the amount of freedom a society offers, and the pace at which interpersonal relationships are disposed, can make it easier or harder for someone to recover from rejection (Sato et al., 2014). Thus, associations might differ in regions where people are less likely to think about rejection (e.g., Industrialized nations). Individual characteristics such as age and sex are also likely to be important. Men and women have different relationship expectations (Buss & Schmitt, 2019) and traits are often found to be more or less important for relationship outcomes in men or women (Allen &

Walter, 2018). People also experience more turbulent relationships in younger adulthood (Lantagne & Furman, 2017) and relationship experience is likely to factor into the importance of rejection sensitivity for relationship outcomes. Therefore, this research synthesis sought to test whether these sociodemographic factors moderate associations between rejection sensitivity and relationship outcomes.

2. Method

The meta-analysis was conducted in line with the PRISMA-2 statement for the reporting of systematic reviews and meta-analyses (Page et al., 2021). The PRISMA-2 checklist is available in Supplementary File S1. The data files for all analyses are available in a permanent online repository (<https://osf.io/2vmbh/>).

2.1. Eligibility criteria

Observational studies assessing rejection sensitivity and romantic relationships were eligible for inclusion. Studies were eligible for inclusion if they included a trait measure of rejection sensitivity and assessed at least one component of romantic relationships. Studies that measured experience of rejection or used tasks to develop a sense of

rejection were not eligible for inclusion. Outcome measures were included provided they assessed experiences relating to a romantic relationship. Abstracts and conference proceedings without a full methods section were not eligible for inclusion.

2.2. Search strategy

A systematic search was conducted in October 2020 using seven electronic databases. The databases used were: Pubmed, Scopus, APAPSYInfo, APAPSYArticles, Elsevier, SAGE, and Web of Science. A broad search was adopted using a single search term: “rejection sensitivity” (see Supplementary File S2). The search string was applied to full texts of articles and was not restricted by language or publication status. A single researcher screened the titles, keywords, and abstracts of all studies identified through the electronic search. Full texts of studies that met eligibility criteria, and studies where the relevance was unclear, were obtained for full text review. The full screening procedure is illustrated in Fig. 1. A total of 4844 records were identified through electronic databases with 2802 unique records. After title, keyword, and abstract screening, 145 articles were extracted for full text search.

Two researchers independently assessed the full texts of all identified studies using Covidence software. In instances where the two

Flow diagram of the screening process

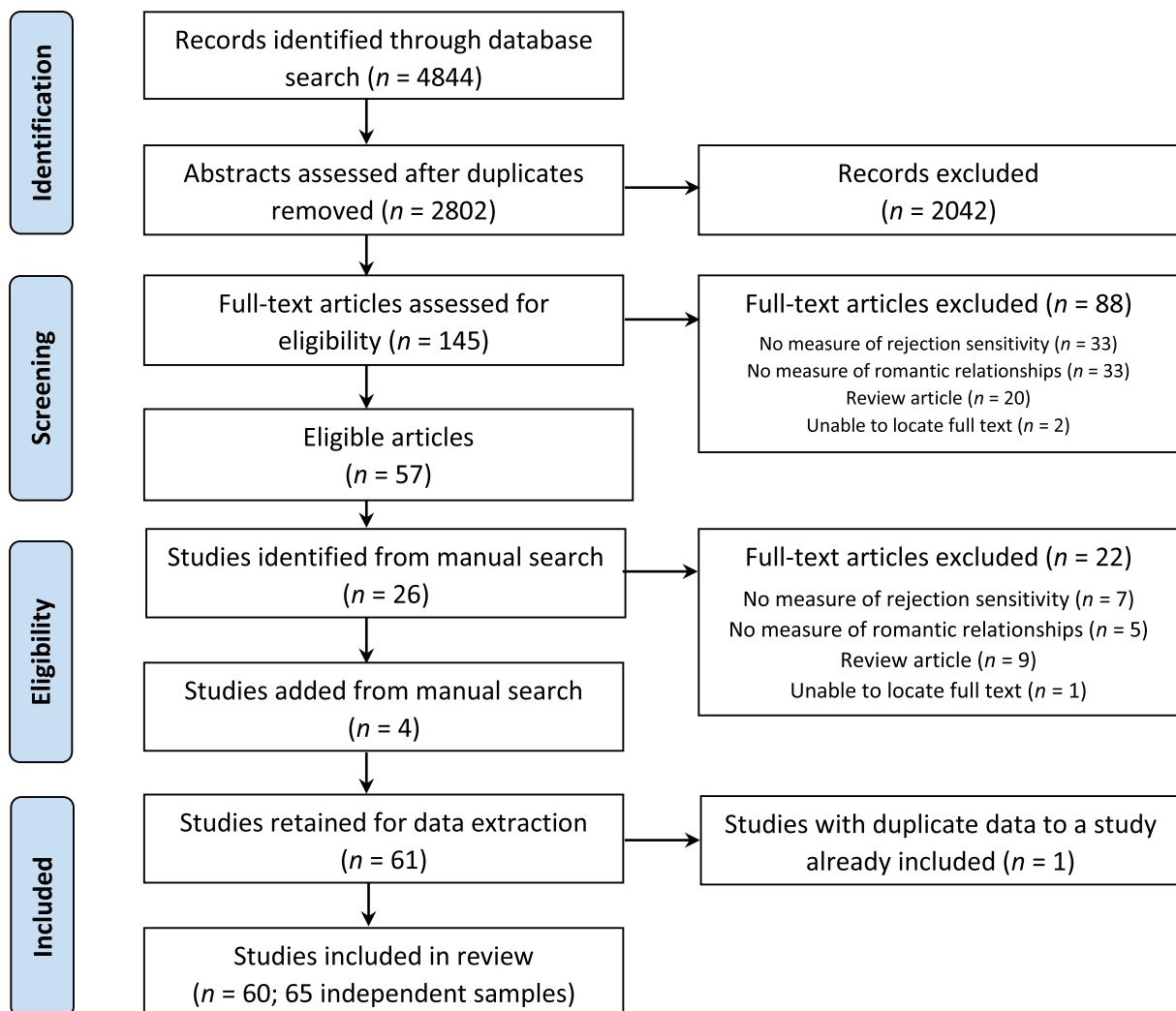


Fig. 1. Flow diagram of the screening process.

researchers disagreed, the suitability of the study was resolved through discussion between the two researchers. In total, 89 articles were excluded for having no measure of rejection sensitivity ($n = 33$), no measure of romantic relationships ($n = 33$), theoretical or review article ($n = 20$), inability to locate full text ($n = 2$), or the same sample used across different studies ($n = 1$). In total, 56 articles were included from the electronic search. A manual search of introduction sections and reference lists of included studies was then conducted using a snowball search strategy. From the manual search, four additional articles were identified that were eligible for inclusion. In total, 60 studies (incorporating 65 independent samples, owing to multi-study articles) were included in the meta-analysis (see Supplementary File S3).

2.3. Data extraction and study quality

Data extraction was performed by a single researcher and was checked by a second researcher with substantial experience in systematic reviews. Data extracted from each article included: author and year, the country where the study took place, study design (prospective or cross-sectional), sample size, the number and percentage of men/women in the study, sample age (mean and standard deviation, or age range if mean not reported), the rejection sensitivity measure used, the relationship variables measured, effect size estimates, and other information used to assess study quality. Study quality was assessed using a critical appraisal tool for assessing research quality – the AXIS tool (Downes et al., 2016). The AXIS tool consists of 20 items that measure aspects of study quality (each scored as 0 or 1) including statement of aims, detailed methods, justification of sample, statistical procedures, discussion of limitations, and reporting of conflicts of interest and ethics. Each study was assigned a score from 0 to 20 with higher scores denoting better reporting standards. Study quality scores were computed for all included studies (Supplementary File S4).

2.4. Meta-analytic strategy

Calculation of pooled mean effect sizes were conducted using inverse-variance weighted random effects meta-analysis. The inverse-variance weighted model assumes that the true effect size varies in each study and assigns a weighting to each effect size based on the inverse of its variance, giving more weight to more precise studies (Borenstein et al., 2009). The effect size metric we use is the zero-order correlation coefficient (r) as this was the most common statistic reported in studies. Effect sizes were taken directly from published studies or were converted to r prior to analyses using standard formulae (Borenstein et al., 2009). In instances where a study reported a standardized regression coefficient but not a correlation coefficient, r was imputed using the following formula: $r = 98\beta + .05\lambda$, where λ is an indicator variable that equals 1 when β is positive and 0 when β is negative (Peterson & Brown, 2005).

In cases where a study reported results without a usable effect size, the authors of the study were contacted and the missing effect size was requested. In total, 12 authors were contacted and 7 authors provided us with the missing effect size. The remaining studies were retained in the meta-analysis but were used in sensitivity analyses only. For sensitivity analyses, in instances where a study reported a non-significant association, but did not report an effect size, the data were first explored with the study excluded and then with an effect size of zero imputed for the study to check on the robustness of results (Pigott, 1994). In instances where a significant positive or negative association was reported, but with no effect size presented, data were first explored with the study excluded and then with the pooled mean effect imputed for the study (Pigott, 1994). In addition, some studies provided multiple effect sizes for conceptually similar constructs (e.g., sexual identity and sexual orientation). In these instances, we extracted the measure that most closely approximated other studies in the analysis or used an averaged effect size from the two measures (see Supplementary File S5 for

details).

For analyses with a sufficient number of effects, a regression asymmetry test was used to identify small sample effects that can indicate publication bias (Egger et al., 1997). If there is no small sample bias then estimates should vary most in small sample studies (owing to random error) and least in large sample studies. Asymmetry in the predicted funnel shape of the plot, alongside a significant regression coefficient ($p < .05$; two-tailed), suggests evidence of publication bias. The potential impact of publication bias was assessed using the trim and fill procedure (Duval & Tweedie, 2000). Trim and fill uses an iterative procedure to impute additional (missing) effects on the negative side of the funnel plot, recomputing the pooled effect size at each iteration until the funnel plot is symmetric. The computed estimate provides an indication of how the true effect size might appear in the absence of publication bias.

The I^2 statistic was computed as a measure of the total variation across studies due to true heterogeneity rather than sampling error (Higgins et al., 2003). For the I^2 statistic, a score of 25 reflects low heterogeneity, 50 represents moderate heterogeneity, and 75 represents high heterogeneity (Higgins & Thompson, 2002). We also report the Q statistic, which provides a test of the hypothesis that variation in effect sizes across studies is greater than that expected by chance. A statistically significant Q statistic alongside an I^2 value above 50 % should prompt the search for potential moderators of the effect. To test for the impact of moderating variables, we used random-effects meta-regression (Borenstein et al., 2009), in which the pooled effect size is set as the criterion variable and the moderating variable is entered as a predictor, with studies weighted by their inverse-variance weights. Tested moderators included sample age, study quality (AXIS score), world-region, and sample sex (percentage of men in the sample). Maximum likelihood estimation was used to calculate regression coefficients. Analyses were computed using *Comprehensive Meta-Analysis 3.0* statistical software (Borenstein et al., 2014). Interpretation of effect sizes were based on current guidelines for effect sizes in psychological science (Funder & Ozer, 2019). An effect size $r = 0.10$ is considered small at the level of single events (but potentially more consequential in the long run), $r = 0.20$ is considered medium (some explanatory and practical use in the short term), and $r = 0.30$ is considered large (impactful in the short term and potentially powerful in the long term) (Funder & Ozer, 2019).

3. Results

The characteristics of included studies are summarized in Supplementary File S6. The 60 articles (65 independent samples) included 147 effect sizes. There were 53 cross-sectional studies and 12 prospective studies, and a total of 16,955 participants. Samples were from North America ($n = 51$), Europe ($n = 8$), Asia ($n = 3$) and Australasia ($n = 3$). Most studies used young and middle age adult samples (grand mean age = 23.46 years, $SD = 7.23$). Study quality scores ranged from 12 to 18 ($M = 15.38$, $SD = 1.59$). Findings from the meta-analysis are presented in Table 1.

3.1. Dating variables

Relationship status showed a small negative association with rejection sensitivity, such that people not in a relationship tended to be more rejection sensitive, $k = 5$, $r = -0.14$ (95 % CI: $-0.22, -0.07$). There was no evidence of publication bias and examination of the Q and I^2 statistics indicated no meaningful heterogeneity. Online dating frequency had a medium positive association with rejection sensitivity, $k = 3$, $r = 0.20$ (95 % CI: $0.14, 0.26$). There was no evidence of publication bias or heterogeneity. For offline dating frequency, there was no significant association with rejection sensitivity, $k = 2$, $r = -0.24$ (95 % CI: $-0.48, 0.04$). However, sensitivity analysis involving the imputation of one additional (negative) effect changed the result to a small significant negative association, $k = 3$, $r = -0.21$ (95 % CI: $-0.32, -0.10$). These effects were such that more rejection sensitive individuals tended to date

Table 1

Random-effects mean associations for rejection sensitivity and romantic relationship variables.

	<i>k</i>	<i>n</i>	<i>r</i> (95 % CI)	<i>Q</i>	<i>I</i> ²	<i>E</i> (<i>p</i>)
<i>Dating variables</i>						
Relationship status	5	1591	−0.14 (−0.22, −0.07)	7.55	46.98	0.50
Online dating frequency	3	1691	0.20 (0.14, 0.26)	3.52	43.11	0.52
Dating frequency	2	229	−0.24 (−0.48, 0.04)	2.16	53.74	–
Times married	1	129	−0.09 (−0.26, 0.08)	–	–	–
Relationship (average) duration	1	110	0.14 (−0.05, 0.32)	–	–	–
Relationship investment	1	148	−0.18 (−0.33, −0.02)	–	–	–
<i>Affective relationship variables</i>						
Relationship satisfaction	16	1926	−0.27 (−0.34, −0.21)	33.73**	55.53	0.84
Perceived partner satisfaction	4	344	−0.29 (−0.43, −0.14)	6.82	56.05	0.17
Relationship commitment	3	535	−0.14 (−0.35, 0.07)	11.60**	82.76	0.56
Relationship closeness	3	1543	−0.76 (−0.95, −0.14)	483.10**	99.59	0.35
Romantic expression	2	342	−0.26 (−0.44, −0.07)	3.47	71.19	–
Romantic partner support	1	110	−0.13 (−0.31, 0.06)	–	–	–
<i>Negative relationship variables</i>						
Intimate partner violence perpetration	13	2343	0.18 (0.11, 0.25)	32.63**	63.22	0.18
Relationship concerns	11	2291	0.42 (0.29, 0.53)	102.35**	90.23	0.01*
Relationship conflict	11	1724	0.22 (0.14, 0.30)	21.74*	54.01	0.36
Intimate partner violence victimisation	9	1268	0.19 (0.12, 0.26)	12.66	36.81	0.10
Self-silencing behaviours	7	1264	0.23 (0.16, 0.29)	10.15	40.91	0.61
Intimate partner violence (combined)	5	641	0.18 (0.05, 0.32)	11.11	64.00	0.17
Power in relationship	3	455	−0.21 (−0.30, −0.12)	0.08	0.00	0.91
Negativity in romantic relationship	2	298	0.36 (0.26, 0.46)	0.60	0.00	–
Jealousy in romantic relationships	2	363	0.30 (0.21, 0.39)	0.01	0.00	–
Withdrawal	1	43	0.28 (−0.02, 0.54)	–	–	–
<i>Sexuality variables</i>						
Risky sexual behaviour	6	1338	0.08 (−0.02, 0.19)	15.83**	68.41	0.16

Table 1 (continued)

	<i>k</i>	<i>n</i>	<i>r</i> (95 % CI)	<i>Q</i>	<i>I</i> ²	<i>E</i> (<i>p</i>)
Sexual activity involvement	3	673	−0.23 (−0.30, −0.15)	1.98	0.00	0.92
Problematic sexuality/compulsivity	3	884	0.10 (0.03, 0.17)	2.14	6.40	0.76
Sexual orientation	2	750	0.15 (0.08, 0.22)	0.33	0.00	–
Sexual submissiveness	1	110	0.19 (0.00, 0.36)	–	–	–
Safer sex self-efficacy	1	63	−0.26 (−0.48, −0.01)	–	–	–
Internalised homonegativity	1	374	0.12 (0.01, 0.22)	–	–	–
Rape myth acceptance	1	94	0.10 (−0.10, 0.30)	–	–	–

Note: *k*, number of pooled effect sizes; *n*, number of pooled participants; CI, confidence interval; *Q*, heterogeneity estimate; *I*², heterogeneity estimate expressed as a percentage; *E*(*p*), significance value for Egger's regression asymmetry test (values below 0.05 indicate meaningful publication bias). **p* < .05, ***p* < .01.

less frequently offline, but more frequently online. Several variables had been explored in a single study and showed that rejection sensitivity was unrelated to number of marriages and average relationship duration, but had a negative association with relationship investment (see Table 1).

3.2. Affective relationship variables

Relationship satisfaction had a significant medium negative association with rejection sensitivity, *k* = 16, *r* = −0.27 (95 % CI: −0.34, −0.21). There was no significant publication bias but evidence of moderate heterogeneity (*I*² = 55.5). Exploration of potential moderators showed that the association between rejection sensitivity and relationship satisfaction was not moderated by sex, *k* = 16, $\chi^2(1) = 2.95$, *p* = .086, *R*² = 0.24, study quality, *k* = 16, $\chi^2(1) = 1.59$, *p* = .207, *R*² = 0.13, or age, *k* = 16, $\chi^2(1) = 2.91$, *p* = .088, *R*² = 0.32. World-region was not sufficiently varied to explore as a moderator. Sensitivity analysis involving imputation of one missing (negative) effect produced identical results. Perceived partner satisfaction also had a negative association with rejection sensitivity, *k* = 4, *r* = −0.29 (95 % CI: −0.43, −0.14). There was no significant publication bias and inspection of *Q* and *I*² statistics indicated no meaningful heterogeneity. Relationship closeness, *k* = 3, *r* = −0.76, (95 % CI: −0.95, −0.14), and romantic expression, *k* = 2, *r* = −0.26 (95 % CI: −0.44, −0.07) had a negative association with rejection sensitivity, whereas relationship commitment and romantic partner support were unrelated to rejection sensitivity.

3.3. Negative relationship variables

Intimate partner violence perpetration had a medium positive association with rejection sensitivity, *k* = 13, *r* = 0.18 (95 % CI: 0.11, 0.25). There was no evidence of publication bias but evidence of moderate heterogeneity (*I*² = 63.2). The association between rejection sensitivity and intimate partner violence perpetration was not moderated by age, *k* = 13, $\chi^2(1) = 0.02$, *p* = .893, *R*² = 0.01, sex, *k* = 13, $\chi^2(1) = 0.10$, *p* = .753, *R*² = 0.00, or study quality, *k* = 13, $\chi^2(1) = 3.21$, *p* = .073, *R*² = 0.30. Sensitivity analyses involving the imputation of one additional (positive) effect produced identical results, *k* = 14, *r* = 0.18 (95 % CI: 0.12, 0.24). Intimate partner violence victimisation also had a medium positive association with rejection sensitivity, *k* = 9, *r* = 0.19 (95 % CI: 0.12, 0.26). There was no evidence of publication bias and a small amount of heterogeneity (*I*² = 36.8). There was a significant moderation by sample age, *k* = 9, $\chi^2(1) = 9.79$, *p* = .002, *R*² = 1.00, such that the

positive association between victimisation and rejection sensitivity decreased as the mean sample age increased. There was no significant moderation by sex, $k = 9$, $\chi^2(1) = 0.12$, $p = .732$, $R^2 = 0.11$, or study quality, $k = 9$, $\chi^2(1) = 0.44$, $p = .506$, $R^2 = 0.11$. Some studies had also combined scores for intimate partner violence perpetration with intimate partner violence victimisation. Rejection sensitivity had a positive association with likelihood of general experience with intimate partner violence, $k = 5$, $r = 0.18$ (95 % CI: 0.05, 0.32).

Relationship concerns had a large positive association with rejection sensitivity, $k = 11$, $r = 0.42$ (95 % CI: 0.29, 0.53). However, there was some evidence of publication bias, $t(11) = 2.98$, $p = .015$. The trim and fill procedure showed two potential missing effects. However, computation of the pooled mean effect with missing effects imputed increased the effect size, $k = 13$, $r_{\text{pseudo}} = 0.47$ (95 % CI: 0.37, 0.59). There was also evidence of substantial heterogeneity ($I^2 = 90.2$). Moderation analysis showed a significant regression model for sample age, $k = 11$, $\chi^2(1) = 13.37$, $p < .001$, $R^2 = 0.66$, such that the positive association between relationship concerns and rejection sensitivity increased as the mean sample age increased. There were no moderation effects for sex, $k = 11$, $\chi^2(1) = 0.00$, $p = .982$, $R^2 = 0.00$, or study quality, $k = 11$, $\chi^2(1) = 0.07$, $p = .798$, $R^2 = 0.00$. Relationship conflict also showed a significant positive association with rejection sensitivity, $k = 11$, $r = 0.22$ (95 % CI: 0.14, 0.30). There was no evidence of publication bias but meaningful heterogeneity ($I^2 = 54.0$). Meta-regression showed that the association between rejection sensitivity and relationship conflict was not moderated by age, $k = 11$, $\chi^2(1) = 1.71$, $p = .191$, $R^2 = 0.40$, sex, $k = 11$, $\chi^2(1) = 3.39$, $p = .066$, $R^2 = 0.34$, or study quality, $k = 11$, $\chi^2(1) = 1.82$, $p = .178$, $R^2 = 0.43$.

Self-silencing behaviours had a medium positive association with rejection sensitivity, $k = 7$, $r = 0.23$ (95 % CI: 0.16, 0.29). There was no evidence of publication bias but meaningful heterogeneity. Moderation analysis showed a significant regression model for sample sex, $k = 7$, $\chi^2(1) = 5.75$, $p = .017$, $R^2 = 1.00$. This effect was such that the association between self-silencing behaviours and rejection sensitivity increased as the percentage of women in the sample increased. In other words, the association between rejection sensitivity and self-silencing behaviour was stronger among women than among men. No moderation effects were observed for age, $k = 7$, $\chi^2(1) = 0.33$, $p = .565$, $R^2 = 0.11$, or study quality, $k = 7$, $\chi^2(1) = 1.47$, $p = .225$, $R^2 = 0.38$. Perceived power in relationships was negatively associated with rejection sensitivity, $k = 3$, $r = -0.21$ (95 % CI: -0.30 , -0.12), such that high rejection sensitive people perceived themselves as having less power in their relationship. Rejection sensitivity also had a positive association with general levels of negativity in relationships, $k = 2$, $r = 0.36$ (95 % CI: 0.26, 0.46), and jealousy, $k = 2$, $r = 0.30$ (95 % CI: 0.21, 0.39).

3.4. Sexuality variables

Risky sexual behaviour was unrelated to rejection sensitivity, $k = 6$, $r = 0.08$ (95 % CI: -0.02 , 0.19). There was no evidence of publication bias but meaningful heterogeneity ($I^2 = 68.4$). Moderation analysis showed that the association between rejection sensitivity and risky sexual behaviour was not moderated by age, $k = 6$, $\chi^2(1) = 1.29$, $p = .257$, $R^2 = 0.45$, sex, $k = 6$, $\chi^2(1) = 0.65$, $p = .421$, $R^2 = 0.08$, or study quality, $k = 6$, $\chi^2(1) = 1.80$, $p = .179$, $R^2 = 0.24$. Sensitivity analysis involving imputation of two missing (positive) effects narrowed the confidence intervals and changed the result such that there was a small positive association between rejection sensitivity and risky sexual behaviour, $k = 8$, $r = 0.07$ (95 % CI: 0.00, 0.15), $p = .047$. Regression models remained unchanged in sensitivity analysis. Rejection sensitivity also had a positive association with hypersexuality (sexual compulsivity), $k = 3$, $r = 0.10$ (95 % CI: 0.03, 0.17) and a negative association with sexual activity involvement, $k = 3$, $r = -0.23$ (95 % CI: -0.30 , -0.15). These effects were such that more rejection sensitive individuals tended to experience more hypersexuality but had less sexual activity involvement in general. Two studies also showed that bisexual men

tended to have higher rejection sensitivity than gay men, $k = 2$, $r = 0.15$ (95 % CI: 0.08, 0.22).

4. Discussion

This systematic review sought to test associations between rejection sensitivity and facets of romantic relationships. Findings showed that rejection sensitivity had a negative association with relationship status, relationship satisfaction, perceived partner satisfaction, relationship closeness, romantic expression, perceived relationship power, and sexual activity involvement. Rejection sensitivity also had a positive association with intimate partner violence perpetration, intimate partner violence victimisation, relationship concerns, relationship conflict, self-silencing behaviours, and jealousy in romantic relationships. More ambiguous findings (supported in sensitivity analyses only) included positive associations for rejection sensitivity with dating frequency and risky sexual behaviour. The positive association for relationship concerns was stronger in samples with older adults, and the positive association for intimate partner violence victimisation was stronger in samples with younger adults. The positive association between rejection sensitivity and self-silencing behaviours was also stronger in samples with a greater percentage of women. Overall, results are generally in line with predictions in the rejection sensitivity model (Romero-Canyas et al., 2010) and provide evidence that rejection sensitivity has an important connection to romantic relationship parameters.

4.1. Main findings

Relationship status was the most frequently tested dating parameter and had a small negative association with rejection sensitivity. This finding might reflect rejection-sensitive individuals tending to have a negatively distorted view of the future of their romantic alliances, leading them to be more threat sensitive and less likely to enter such romantic relationships (Downey & Feldman, 1996; Pietrzak et al., 2005). This tendency might also explain the finding that rejection sensitive individuals tend to date less frequently. Contrary to expectations, the meta-analysis also showed that rejection sensitive individuals tend to date more frequently online. This finding might be explained by online dating websites providing a less confrontational and more private space that better accommodates anxious expectations of rejection-sensitive individuals. Indeed, research has found that online dating involves fewer non-verbal cues (Lieberman & Schroeder, 2020) that might leave less room for misinterpretation of benign cues as indicative of rejection. It has also been found that more rejection sensitive individuals feel more comfortable in disclosing their real self in online compared to offline environments (Blackhart et al., 2021).

A greater amount of research had explored associations between rejection sensitivity and affective outcomes. Consistent with hypotheses, rejection sensitive individuals tended to have lower relationship satisfaction, lower perceived partner satisfaction, and lower levels of relationship closeness and romantic expression. This finding is consistent with previous meta-analytic findings on the broader trait of neuroticism (Malouff et al., 2010), as well as the rejection sensitivity model that predicts rejection sensitive individuals are more likely to form anxious attachment bonds (Pietrzak et al., 2005). These anxious attachment bonds typically manifest in negative interpretations of the actions of partners, worry about partner's commitment, and overestimates of threats, that can contribute to lower satisfaction, closeness and commitment (Pietrzak et al., 2005). Interestingly, rejection sensitive individuals did not appear to differ in terms of their commitment in romantic relationships. Despite having lower levels of satisfaction, rejection sensitive persons appear as committed to their relationships as less rejection sensitive persons.

The largest body of research had focused on negative outcomes. Relationship conflict had a medium positive association with rejection sensitivity, and relationship concerns had a large positive association

with rejection sensitivity. This finding is consistent with the rejection sensitivity model that describes the process of ready perceptions, in which instinctually generated thoughts increase readiness to perceive rejection and predispose rejection-sensitive individuals to be hyper-attentive to their partner's negative behaviour contributing to greater concern and conflict. For relationship concerns, we also found that the association was stronger in older samples. This finding indicates that rejection sensitivity is less important (in terms of formulating concerns about the relationship) among younger people. This could possibly be because younger adults have grown accustomed to receiving rejection (e.g., from universities, job profiles, other romantic partners) due to the unstable and flexible time of their lives (see Elder, 1998). In addition, older adults have often achieved their desired social and financial standing in society (Singer et al., 2007), which might make them less accustomed to receiving rejection.

Rejection sensitivity also had a positive association with self-silencing behaviours and a negative association with perceived power (such that more rejection sensitive persons tended to perceive themselves as having less power in the relationship). These findings are consistent with the rejection sensitivity model that predicts rejection-sensitive individuals feel less able to voice their concerns in a relationship as doing so could potentially lead to rejection from their partner and dissolution of their relationship (Pietrzak et al., 2005). For self-silencing behaviour, the relationship became stronger as the percentage of women in the sample increased, indicating that rejection sensitive women, more so than rejection sensitive men, are more likely to silence themselves. This finding might be explained by societal expectations that glorify sacrifice among women, or a lack of knowledge about helpful resources (Rizo & Macy, 2011). In addition, rejection sensitive individuals also showed a greater propensity to experience jealousy. This finding might be explained by rejection sensitive individuals having more insecure attachment styles, which can often lead to jealousy of persons around their partners due to their heightened urge to prevent rejection (Martínez-León et al., 2017).

An important finding was that rejection sensitivity had a medium positive association with intimate partner violence perpetration. This finding is consistent with a previous meta-analysis (not exclusive to romantic relationships) that found rejection sensitive individuals tend to display greater levels of aggression in general (Gao et al., 2021). This greater propensity towards violence might reflect rejection sensitive individuals tending to overthink and negatively interpret their partner's intentions and behaviour, which can generate frustration and intense reactions, expressed through perpetrating violence on their partner (Romero-Canyas et al., 2010). We also found that rejection sensitivity had a medium positive association with intimate partner violence victimisation. This finding is also consistent with a previous meta-analysis that found rejection sensitive individuals tend to experience more victimisation in general (Gao et al., 2021). Importantly, we also found that the association decreased as the sample age increased. This indicates that rejection sensitivity is more important for victimisation among younger adults and might be explained by younger adults having less access to tangible and informational resources to manage unhealthy relationships.

The final category of relationship variables included components of sexuality. We found that rejection sensitivity was unrelated to risky sexual behaviour (typically measured as unprotected sex with a new sexual partner). However, after inclusion of two missing effect sizes (estimated at the pooled mean effect from other studies) we found that rejection sensitivity had a small positive association with risky sexual behaviour. This finding aligns with a recent meta-analysis that found a positive association between anxious attachment style and risky sexual behaviour (Kim & Miller, 2020). That is, the tendency for rejection sensitive persons to experience anxious attachments might explain the finding that rejection sensitive persons are more likely to engage in risky sexual behaviour. In addition, sexual activity constitutes a high-risk situation for potential rejection and rejection sensitive individuals

tend not to discuss matters with their partner that might pose a threat to their relationship (Downey et al., 1999). This might also explain the finding that more rejection sensitive individuals tended to have less sexual activity involvement in general. In other words, due to their inability to communicate about healthy sexual practices, rejection sensitive individuals either find themselves not engaging in sexual activities or engaging in risky sexual activities. In addition, more rejection sensitive individuals might have less frequent sexual involvement due to the delayed onset or lower likelihood of entering a romantic relationship (Downey et al., 2000).

Rejection sensitivity was also found to relate to sexual orientation such that men identifying as bisexual tended to be more rejection sensitive than men identifying as exclusively gay. This finding differs somewhat from a recent meta-analysis that explored sexual orientation and the broader trait of neuroticism (Allen & Robson, 2020), that found no difference between gay men and bisexual men, but that bisexual women were higher in neuroticism than lesbian women (Allen & Robson, 2020). This finding might indicate that rejection sensitivity is a more important trait for sexual identity development among men. However, with only two effect sizes available, and no heterosexual comparison group, more research is required to explore rejection sensitivity across sexual orientations. A final finding from the current meta-analysis was that rejection sensitive individuals were more prone to hypersexuality. This finding is consistent with a previous meta-analysis on the broader trait of neuroticism that found a large positive association between neuroticism and hypersexuality (Allen & Walter, 2018). Neuroticism is thought to relate to hypersexuality because people high in neuroticism have difficulty managing thoughts which makes it difficult for such individuals to control their intense preoccupation with sexual thoughts and urges (Allen & Walter, 2018).

4.2. Limitations

Strengths of this research include the systematic approach to study identification, the use of quantitative techniques to test hypotheses, and the inclusion of sensitivity analyses and moderation analyses to more fully explore and interpret results. However, there are some important limitations that readers must consider when interpreting findings. First, most samples were from North America and there was an underrepresentation of participants from Asia, Europe, Australasia, and Africa. The nature of romantic relationships is thought to be contingent on culture and local practices (Sato et al., 2014) and it is unknown whether findings will transfer to all world regions. Second, sample age and sex were tested as potential moderators where possible, with an indication that some associations are stronger among younger/older adults or women. However, it was not possible to test moderators in all cases and in instances where moderator analyses were run the data only had sufficient statistical power to detect very large effects. In other words, inability to detect a moderation effect in meta-regression could indicate that there is no effect to be found, but could also reflect the small number of studies and corresponding low statistical power (Borenstein et al., 2009). Tied into the point above, a third limitation is that some associations had only been tested in one or two (small sample) studies, meaning we can have less confidence in the effect sizes emerging in those analyses.

A fourth potential limitation is that studies tended to assess variables using self-report measures. Self-report measures are subject to social desirability bias and this might contribute to some attenuation of effect sizes. Relationship variables are particularly sensitive so social desirability bias and future research might look to incorporate additional (partner-report) measures for a more rounded interpretation of results. A fifth limitation is that most studies used cross-sectional designs. Future research should consider prospective analyses that test how change in rejection sensitivity over time corresponds with change in relationship parameters over time. A final limitation is that the majority of studies examined outcomes at the level of the individual rather than that of the couple. Couples often differ in their perceptions of the same constructs,

such as the amount of work done by each person (e.g., housework), or how much each person pays for shared resources. Researchers should consider assessments that test how rejection sensitivity of one individual corresponds to outcomes measured by both individuals.

5. Conclusion

Improving the quality of romantic relationships requires an understanding of the various factors involved in shaping romantic experiences. This study has found that more rejection sensitive individuals reported lower levels of relationship satisfaction, relationship closeness and perceived partner satisfaction, and higher levels of relationship concerns, relationship conflict, intimate partner violence, self-silencing behaviours, and jealousy in romantic relationships. There was also evidence that rejection sensitive individuals are more likely to engage in risky sexual behaviour and are more prone to sexual compulsivity. These findings might be of interest to relationship and marriage counsellors and other relationship professionals. For example, rejection sensitivity assessments might offer a useful screening tool to identify individuals who might benefit most from participation in interventions aimed at improving romantic relationships. Increasing awareness of associations might also help individuals (couples) high in rejection sensitivity to identify their own behavioural tendencies and how they might impact relationship quality. We encourage continued research of rejection sensitivity in romantic relationships, particularly in underrepresented world-regions, to ultimately improve interpersonal care and promote healthy relationships in society.

Declaration of competing interest

The authors declare no competing interests.
There were no sponsors or funders for the review.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.paid.2023.112186>.

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